Energy Strategy Reviews 61 (2025) 101816

Contents lists available at ScienceDirect

Energy Strategy Reviews

journal homepage: www.elsevier.com/locate/esr

Efficiency enhancement of energy supply chains using a machine learning-driven network evaluation framework for blockchain adoption

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ARTICLE INFO

Handling Editor: Dr. Mark Howells

Keywords:
Blockchain adoption
Decision-making
Network data envelopment analysis
Energy industry
Oil and gas supply chain

ABSTRACT

The Oil and Gas (O&G) sector is essential to the global energy supply, yet it still struggles with inefficiencies caused by conventional, paper-driven decision-making processes in its supply chain. This study aims to assess the integration of blockchain technology as a strategic solution to these challenges. It examines strategies for blockchain integration considering organizational, supply chain, technological, and external factors. A novel approach based on four network-based data-driven optimization models is utilized to investigate a case study in Norway's O&G sector over a five-year period (April to October 2020). These models incorporate uncertainty to assess the effectiveness of various blockchain integration strategies. The findings demonstrate that "single-use strategy" is effective in low-risk scenarios, while "substitution strategy" is optimal in high-risk scenarios. The application of the Super-Efficiency Network Data Envelopment Analysis (SENDEA) model represents its robustness in evaluating blockchain adoption within the industry. Furthermore, a an efficient Machine Learning (ML) technique; i.e., Principal Component Analysis (PCA), is utilized to further assess blockchain adoption strategies. Therefore, policymakers are recommended to support risk-based blockchain adoption by directing organizations to implement single-use strategies in low-risk environments and substitution strategies in high-risk circumstances. Additionally, they can take advantage of practical evaluation frameworks provided by the SENDEA and uncertain models to systematically appraise and prioritize blockchain strategies based on their potential impacts.

1. Introduction

Affordable and sustainable energy serves as the basis for development in key areas such as agriculture, education, healthcare, and transportation [1]. Although progress toward global energy goals has been consistent, it remains insufficient. Overcoming this challenge requires significant investment in renewable energy infrastructure; e.g., solar, wind, and thermal systems, integrated with advancements in technology and grid modernization, particularly in developing countries. Achieving Sustainable Development Goal 7 (SDG 7: guaranteeing affordable, reliable, sustainable, and modern energy access for everyone) requires countries to accelerate electrification, increase investment in renewable energy, enhance energy efficiency standards, and implement supportive policies. The involvement of the private sector plays a crucial role, with enterprises and investors promoting sustainable innovations and practices. Deploying blockchain solutions can facilitate faster achievement of SDG 7 b y improving transparency, operational efficiency, and confidence in the energy industry. For

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https://doi.org/10.1016/j.esr.2025.101816
Received 22 June 2024; Received in revised form 2 June 2025; Accepted 31 July 2025
Available online 18 August 2025
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example, Nikzad and Mehregan [2] presented a relevant renewable-based integrated solution by utilizing a solar-powered photovoltaic system to sustainably support blockchain adoptions like Ethereum mining. They built up a practical model combining clean energy with decentralized digital economies, significantly enhancing economic viability and reducing environmental impact.

The 27th United Nations Climate Change Conference (COP27) was held in Sharm El-Sheikh, Egypt, from 6 to November 20, 2022 under the theme "Together for Implementation" and chaired by Egypt on behalf of the African region [3]. Almost 35,000 delegates from 198 countries convened to fast-track action under the Paris Agreement. While COP26 focused on setting targets, COP27 prioritized implementation, although advances in emissions reduction were still insufficient. Despite urging for revised emissions targets, only 29 nations put forward stronger Nationally Determined Contributions (NDCs), with global temperatures still forecasted to rise by 2.4 °C. UN Secretary-General António Guterres has called on global leaders to take meaningful action, cautioning that present efforts are far from meeting the 1.5 °C target. Blockchain can support the implementation focus of COP27 by enhancing transparency, accountability, and traceability in climate actions.

Behera et al. [4] examined the challenge of balancing economic growth and ecological sustainability in BRICS nations (i.e., Brazil, Russia, India, China, and South Africa). They demonstrated that while hydro and nuclear energy use, along with political stability, increases the ecological footprint, green technology innovation significantly reduces it. Moreover, they recommended coordinated policies to enhance regulatory frameworks, promote technological innovation, and facilitate clean energy adoption for sustainable development. In another study, Behera et al. [5] analyzed the impact of renewable (hydro and nuclear) and non-renewable (coal and oil) energy consumption on India's economic growth from 1985 to 2021. The findings showed that while hydro and oil consumption impede growth, nuclear and coal energy augment long-term economic growth. They recommended investing in renewable energy to balance India's energy mix for sustainable growth. It was shown that blockchain can enhance transparency, accountability, and real-time monitoring of energy use, supporting clean energy adoption and policy coordination for sustainable growth in BRICS nations.

Investigating blockchain integration within O&G operations is essential to advancing SDG 7. This technology improves transparency across the supply chain, increases operational efficiency, and builds trust among stakeholders. The immutable data records of blockchain enable precise monitoring of emissions, aiding energy companies in aligning with the Paris Agreement and adhering to regulatory standards. In addition, by creating transparent and verifiable environmental records, blockchain can fundamentally change how decarbonization efforts are tracked and validated across the sector.

1.1. Blockchain technology and blockchain in the O&G sector

In the last decade, blockchain technology has become a pervasive trend in energy industries due to its vast benefits [6,7]. For example, in responding to the O&G sector's digitalization, major companies explore blockchain's potential benefits in management, efficiency, and data security [8]. According to Lu et al. [8], Europe and Asia spearhead blockchain development in O&G, yet operational projects are globally limited. Despite opportunities like reduced transaction costs and improved transparency, challenges endure, including industry understanding gaps, experimental applications, and inadequate investment. Blockchain transforms the data-intensive O&G industry by enhancing efficiency in administrative functions and addressing challenges like price volatility. Lakhanpal & Samuel [9] highlighted its benefits in the supply chain, regulatory compliance, database management, and cybersecurity and enabled industry stakeholders to better understand its potential to transform transparency, and enhance security and efficiency. Existing systems in O&G Supply Chain Management (SCM) lack adequate transparency, security, and traceability. Our study investigates the capabilities of blockchain technology to offer decentralized, secure, and transparent solutions.

Ahmad et al. [10] comprehensively examined state-of-the-art blockchain schemes, initiatives, and case studies, illustrating its practical applicability in automating pivotal services, including product tracking, trade document protection, and the coordination of purchasing activities within the O&G industry. Blockchain technology, characterized by its decentralized, secure, and transparent architecture, has the potential to revolutionize sectors including finance, SCM, healthcare, and education. In the context of education, the integration of blockchain offers opportunities for enhancing efficiency, security, and credibility through the establishment of secure platforms facilitating the tracking of academic achievements [11]. It enhances the functionality of digital cryptocurrency, e.g., Bitcoins [12]. Blockchains are digital public ledgers recording all Bitcoin transactions [13], visible to every member of the network, ensuring transparency. The structure makes data manipulation difficult due to the hash values in each block, creating a secure environment. As blockchain technology demonstrates success in numerous cryptocurrencies, diverse organizations seek to utilize its transparency and fault tolerance to address challenges involving multiple untrusted actors in resource distribution.

1.2. Blockchain impact on SCM

Amiri et al. [14] proposed an innovative blockchain system design to enhance the supply chain of Engineering, Procurement, and Construction (EPC) companies in O&G infrastructure development, addressing cost and time inefficiencies. Their findings revealed significant reductions in cost inefficiencies (12.4%) and operation lead times (36.5%). While limited by the confidentiality of the EPC company's information, this research introduces a novel blockchain solution for project-development supply chains, contributing to improved efficiency and implementation processes. British Petroleum demonstrated that oil and natural gas collectively constitute 57% of the total energy consumed, with a projected expansion of 1.8% in global oil consumption and a 3% increase in natural gas consumption. Despite significant attention toward alternative energy sources, it is anticipated that O&G will maintain a share of over 50% in the global energy industry by 2040 [15,16]. It is important for supply chains to maintain product quality during shipment. Traditional centralized management systems are vulnerable to corruption and misuse of authority. Blockchain provides a distributed solution, enhancing transparency and visibility [17]. The globalization of supply chains presents management challenges which can be efficiently treated by digitalization and improving resilience [18]. Blockchain technology, examined by Saberi et al. [19], is known as a potential to improve transparency, traceability, and security in global supply chains. They explored blockchain's potential applications in SCM and investigated its role in supporting sustainability. They also identified potential adoption barriers, including inter-organizational, intra--organizational, technical, and external factors.

Each action in the supply chain, facilitated by these digital technologies, is systematically recorded on the blockchain [20]. This blockchain acts as an immutable repository for information universally accepted by all participating parties. Blockchain technology ensures confidentiality, integrity, and availability of transactions across a distributed ledger, revolutionizing various industries. Its flexible architecture accommodates a wide range of industry requirements, tackling key challenges related to usability, security, and cost. Built-in security features maintain the integrity and transparency of supply chain transactions and underscore the technology's transformative potential [21]. Numerous studies have emphasized blockchain's potential to revolutionize traditional supply chain practices, particularly in mitigating trust issues and improving visibility. With its shared ledger, blockchain improves information transmission timeliness and diminishes costs associated with searching and negotiating. Smart contracts further streamline operations, enhancing interactions among supply chain

participants. Yet, despite its transformative potential, discussions on blockchain-based business models remain limited, with a critical need for comprehensive evaluation by supply chain practitioners [22]. Traceability plays a key role in tracking information and contributes significantly to supply chain transparency. As a decentralized ledger technology, blockchain resolves the limitations of centralized traceability methods and ensures records are securely maintained. Integrating it into supply chains augments transparency through the tracking of events.

1.3. Research objectives

Accordingly, the research objectives are also defined as follows.

- o Evaluating strategies for using blockchain specifically in the O&G industry,
- o Introducing interrelated criteria for evaluating blockchain application strategies based on supply chain, technology, drivers, and organization,
- o Creating a network evaluation model to analyze the relationships between criteria in evaluating blockchain strategies,
- o Employing Network Data Envelopment Analysis (NDEA) models to evaluate the effectiveness of blockchain deployment strategies,
- o Integrating reality-based conditions, including uncertainty, into the evaluation models for a more comprehensive analysis of blockchain deployment strategies.

1.4. Research structure

This study introduces a novel methodology for evaluating blockchain implementation strategies in the O&G supply chain. Unlike previous work that has largely overlooked blockchain adoption in this sector, this research provides a comprehensive evaluation framework fine-tuned to align with the O&G industry's priorities in terms of transparency, efficiency, and resilience. Aligned with its objectives, to analyze blockchain strategies, establish interconnected evaluation criteria, and account for real-world uncertainties, the study combines data-driven and optimization-based techniques using NDEA. A key innovation is the development of models that not only reflect the complex relationships among evaluation factors but also assess risk under both stochastic and deterministic scenarios. We explore the interrelationships among evaluation criteria across various dimensions, including supply chain, technology, and organizational drivers. Principal Component Analysis (PCA) method is then utilized to transform optimal data into a simpler, more interpretable structure while preserving the most important patterns of variation. As a result, it drives forward strategic decision-making in the adoption of blockchain technology, offering O&G supply chain managers a more accurate and practical tool for managing transformation in the energy sector.

The remaining sections are organized as follows. A literature survey is performed in Section 2 in order to highlight the gaps and contributions. Section 3 states the problem and describes the models. The obtained results are presented in Section 4 which are then discussed in Section 5. Finally, Section 6 concludes the research and provides an outlook for future work.

2. Literature survey

2.1. Blockchain adoption in SCM

Xu et al. [23] investigated blockchain technology adoption in the automotive industry, highlighting its potential benefits for German OEMs, such as improved product information aggregation and supply chain reliability, while identifying obstacles such as technology immaturity and lack of standards. Waqar et al. [24] examined blockchain technology's impact on construction supply chains, revealing its potential to enhance transparency, traceability, and efficiency through empirical case studies and industry surveys. Chiaraluce et al. [25] analyzed blockchain adoption in agri-food supply chains, particularly in wine and olive oil, highlighting its potential to enhance traceability, safety, and sustainability while addressing fragmented research. Amico and Cigolini [26] evaluated blockchain technology's impact on port logistics, specifically the bill of lading lifecycle. Their simulation study revealed blockchain enhances trust, security, and efficiency, reducing container lead time by 3–4%. While the significance of employing blockchain in the supply chain, particularly within the O&G industry, is paramount, there is a paucity of literature offering optimization and data-driven models about its utilization in this domain. Babaei et al. [27] integrated blockchain transparency into a tri-level Supply Chain Network (SCN) using a bi-objective optimization model. They utilized fuzzy Goal Programming (GP) and chance-constrained programming to respectively deal with the bi-objectiveness and uncertainty of the model. The improved branch and efficiency algorithm assessed transparency, cost, and service. Comparing cases emphasized the trade-off between blockchain costs and benefits in supply chain design. Meeting customer demand is a growing concern for supply chain managers. The omni-channel strategy addressed this by integrating online and traditional channels.

Babaei et al. [28] and Babaei et al. [29] suggested an algorithm for assessing omni-channel distribution network configurations using an extended Data Envelopment Analysis (DEA) model. The algorithm evaluated the network configurations based on cost, service, blockchain transparency, and environmental criteria, employing an uncertain optimization model for consensus. As economies of scale drive demand growth, Babaei et al. [30] proposed a sustainable, customer-centric closed-loop SCN design integrating blockchain technology for transparency, as a first-of-its-kind design, applying fuzzy GP in iterative and integrated DEA models. Babaei et al. [31] offered a bi-level multi--objective DEA-based optimization for production and distribution planning, tackling cost and traffic congestion, and incorporating blockchain transparency. Solutions, categorized by strategies, emphasized either cost or traffic congestion efficiency. In addressing environmental challenges and the dominance of road transportation, Oudani [32] explored synchromodal transportation with blockchain technology. A multi-objective optimization model integrated transportation and blockchain costs for sustainable solutions. With an emphasis on blockchain's essential function, especially in sectors dealing with hazardous materials, Oudani et al. [33] proposed a green blockchain system. Mathematical models optimized energy management for cost-effective and eco-friendly blockchain transactions, ensuring security and sustainability. Babaei et al. [34] developed a decision support framework using DEA models to evaluate blockchain adoption strategies in the O&G supply chain, addressing transparency and efficiency issues. They also considered Norway's O&G sector and represented varying outcomes under deterministic and uncertain conditions, offering a comprehensive approach to guide strategic decision-making. De Carvalho et al. [35] suggested an approach to optimize the adoption of blockchain in a SCN configuration. Despite blockchain's potential to improve transparency, efficiency, cost savings, and traceability, its adoption was limited. A Multi-Criteria Decision-Making (MCDM) procedure was utilized by Munim et al. [36] for appraising blockchain implementation strategies in the O&G industry. Factors like a lack of expertise and collaboration significantly impacted the adoption process. The Bayesian Best-Worst Method (BWM) identified transformation as the leading strategy for accelerating blockchain adoption.

Distributed ledger technologies like blockchain hold the potential for enhancing SCM and sustainability, promoting responsible sourcing and environmental compliance. Despite its potential, blockchain faces limited adoption due to managerial unfamiliarity. Yousefi & Tosarkani [37] employed network theory, fuzzy inference system, fuzzy cognitive map, and hybrid learning algorithm to find and rank blockchain adoption enablers, demonstrating its remarkable effect on mineral supply

chain performance. Since 2008, academia has increasingly focused on blockchain technology, particularly its application in sustainable manufacturing. However, technical challenges persist, such as consensus algorithms and privacy protection. To address these, a hybrid fuzzy MCDM framework was proposed by Su et al. [38], evaluating criteria weights and organizational preferences. An empirical case study underscores the framework's effectiveness, highlighting blockchain's potential in transparency, trust establishment, and fraud prevention. Yet, further research on blockchain adoption in sustainable manufacturing is warranted. Babaei et al. [7] investigated how blockchain technology could transform SCM by enhancing transparency, despite environmental and security challenges. They applied a tri-objective optimization model for designing supply chain-blockchain technology networks, considering energy consumption, emissions, and security.

Table 1 presents a summary of the existing literature on blockchain technology and its applications across various supply chain industries. This table highlights key studies, their applications, main focus, and findings, along with two important conditions that demonstrate stochasticity and calculate risk based on uncertainty and super-efficiency.

2.2. Nexus between blockchain adoption drivers and supply chain outcomes

The current body of literature consistently highlights a causal link between the drivers of blockchain adoption and improvements in supply chain performance. Across various sectors, factors such as technological preparedness and the implementation of strategic approaches are specified to significantly enhance transparency and operational effectiveness. However, within the O&G industry, this connection remains insufficiently examined. Preliminary research, including Munim et al. [36], Babaei et al. [39], and Babaei et al. [34], points to organizational, technological, and supply chain-related elements, along with stochastic influences, as key moderators of blockchain's effectiveness. Despite these insights, there is a notable gap; existing studies have not comprehensively assessed how interconnected decision-making criteria, under conditions of uncertainty and from the perspectives of super-efficiency and risk, shape the success of blockchain strategies in real-world O&G supply chain environments. This study addresses that gap by introducing a novel NDEA framework designed for such complex evaluations.

2.3. Research gaps and contributions

The main research gaps include.

- Limited research on the evaluation of blockchain strategies in the supply chain, particularly in industries like O&G,
- Absence of studies examining the relationships between criteria in evaluating blockchain strategies,
- Lack of network evaluation models for assessing blockchain application strategies,
- Scarcity of models combining both data-based and optimization-based approaches for evaluating blockchain deployment strategies,
- Insufficient consideration of reality-based conditions, such as uncertainty and discriminating power, in the existing literature on blockchain strategy evaluation.

Based on the above-mentioned research gaps, the contributions of this work are listed as follows.

Table 1
Overview of the blockchain adoption studies in supply chain industries.

References	Applications	Focus area	Key Findings	Stochastic Optimization Model	Risk-Based Aggregation Method	
					Uncertainty	Super-efficiency
Yousefi & Tosarkani [37]	Mineral supply chain	Blockchain enablers using fuzzy inference and cognitive maps	- Showcased remarkable influence on mineral supply chain performance	No	No	No
Munim et al. [36]	O&G	Decision-making for blockchain implementation strategies	- Identified key barriers to blockchain adoption and specified transformation as a preferred strategy	No	No	No
Oudani [32]	Transportation	Synchromodal transportation with blockchain integration	- Made a balance between transportation and blockchain costs	No	No	No
Oudani et al. [33]	Hazardous materials supply chain	Green blockchain framework for eco-friendly transactions	- Optimized energy management for secure and cost-effective transactions	No	No	No
Babaei et al. [27]	SCN	Bi-objective optimization with blockchain transparency	- Provided trade-offs between cost, transparency, and service	No	No	No
Babaei et al. [30]	Closed-loop Supply Chain	Sustainable customer-centric blockchain-integrated network	- Proposed a closed-loop network model, emphasizing transparency and sustainability	No	No	No
Su et al. [38]	Sustainable manufacturing	Blockchain adoption criteria using hybrid MCDM framework	- Emphasized blockchain's potential for transparency, trust, and fraud prevention	No	No	No
Xu et al. [23]	Automotive	Blockchain adoption benefits for German OEMs	- Identified potential benefits such as improved product info and supply chain reliability	No	No	No
Waqar et al. [24]	Construction supply chains	Blockchain's impact on transparency and efficiency	- Enhanced transparency, traceability, and efficiency based on empirical evidence	No	No	No
Chiaraluce et al. [25]	Agri-food supply chains	Blockchain's role in traceability and safety	- Noted improvements in safety, sustainability, and traceability, especially for wine and olive oil	No	No	No
Amico & Cigolini [26]	Port logistics	Blockchain's impact on bill of lading	- Improved trust, security, and efficiency - Reduced container lead time by 3–4 %	No	No	No
Babaei et al. [39]	O&G	Examination of blockchain strategies within supply chain frameworks	- Demonstrated that adopting a single-use strategy yields significant efficiency improvements	Yes	No	No
Current Study	**O&G Supply Chains**	**Comprehensive framework for evaluating blockchain strategies**	**- Delineated practices in the O&G sector - Fostering efficiency, transparency, and adaptability through cutting-edge technological solutions and offering novel business models**	**Yes**	**Yes**	**Yes**

Table 2
Strategies and critical criteria influencing the blockchain implementation.

Strategy		Description
Single-use		Low novelty and coordination, suitable for small-scale trials.
Localization		High novelty and low coordination, streamlining local networks and enhancing transparency.
Substitution		High coordination, relies on established applications, with caution against potential challenges.
Transformation		High coordination and novelty, signify a significant shift in systems, requiring changes in social, legal, and political realms.
Main Criteria	**Sub-criteria**	**Description**
Motivators and Barriers	Collaboration Pressure	Necessity for collective blockchain implementation, fostering heightened collaboration for optimal technology utilization.
	Customer Pressure	Customer demand for sustainable products propels blockchain adoption, ensuring credibility and validity within the intricate web of the supply chain.
	Market Pressure	Market dynamics, especially in sectors like O&G, propel organizations to embrace blockchain for sustainability, mitigating emissions, and aligning with evolving expectations.
Drivers	Security Enhancement	Blockchain's temporal sequencing and consensus protocols amplify information security, circumscribing access to digital assets and fortifying secure transactions.
	Operational Cost Reduction	Decentralized blockchain systems optimize operations, markedly curtailing costs, particularly in audits and the establishment of standardized systems.
	Traceability Improvement	Blockchain's inherent tracking capabilities elevate traceability and transparency, conferring a competitive advantage and addressing consumer sustainability considerations.
Organizational Barriers	Lack of Expertise in Technology	Scarcity of competencies and comprehension impedes proliferation, constraining the widespread assimilation of fundamental blockchain principles.
	Lack of Tools for Implementation	Insufficient hardware and software tools lead to challenges, necessitating substantial investments for the effective implementation of blockchain technology.
	Lack of Benchmarking Data	The dearth of benchmarks and standards impedes technological juxtaposition and evolution within the blockchain domain.
Supply Chain Barriers	Lack of Customer Awareness	Inadequate customer understanding and suboptimal communication act as impediments to blockchain adoption, hindering the dissemination of this transformative technology.
	Lack of Partner Collaboration	Partners' hesitancy to share information hampers the transparent deployment of blockchain technology within SCNs.
	Lack of Partner Coordination	Inefficient information sharing and divergent priorities act as constraints, obstructing the efficient adoption of blockchain within the supply chain framework.
Technological Barriers	Immaturity of the Technology	Blockchain's nascent stage introduces challenges in scalability, usability, and interoperability, impinging on its integration within broader technological ecosystems.
	Limited Information Technology Infrastructure	Infrastructure constraints, including cost implications, computational capacity, and transaction volume, impact the implementation landscape of blockchain technology.
	Security Concerns	Despite blockchain's reputation for security, apprehensions arise from vulnerabilities related to third-party involvement and the potential for illicit activities.

I. This work introduces a comprehensive framework for evaluating blockchain strategies in the O&G supply chain, addressing the limited research in this vital industry and providing key insights for performance evaluation and decision-making,
II. It explores the relationships between evaluation criteria in blockchain strategies, an area with limited research. It offers insights into their interconnections and provides a deeper understanding of blockchain strategy evaluation,
III. It addresses gaps in stochastic risky network evaluation models for blockchain strategies by presenting innovative models for assessing blockchain networks in supply chains. We introduce a hybrid framework that combines data-based and optimization-based approaches and incorporates real-world factors for a more accurate evaluation of blockchain deployment strategies. This approach also derives risk from evaluation results under both stochastic and deterministic conditions, which is novel in this context. Furthermore, this study utilizes PCA technique based on optimal evaluation outcomes. It enables managers to benefit from Machine Learning (ML)-driven insights from both methodologies and strategies viewpoints.

It should be noted that the investigation of blockchain strategies in the O&G sector is very limited. The three notable studies in this area are Munim et al. [36], Babaei et al. [39] and Babaei et al. [34]. The main novelties here is that the optimization models are developed based on linear programming and the focus is on analyzing stochastic conditions rather than incorporating very specific uncertainty conditions. Additionally, the suggested approach calculates risk under both stochastic and deterministic conditions and derives this risk from different levels of discrimination power in evaluation models, such as traditional and super-efficiency models. In addition, the current study extends the perspective of Babaei et al. [34] by advancing network data analysis, an aspect overlooked in their work.

2.4. Implementation steps of the model

The implementation steps of the model are depicted in Fig. 1 which are also described as follows.

1. **Identifying Strategies:** Define blockchain deployment strategies, focusing on factors like application novelty and coordination, as summarized in Table 2,
2. **Establishing Evaluation Criteria:** Compile criteria categorized under supply chain, organization, technology, drivers, and pressures, detailed in Table 2,
3. **Developing Evaluation Models:** Use various NDEA models, incorporating real-world conditions such as uncertainty and discrimination power (Fig. 8),
4. **Analyzing Results and Aggregation:** Evaluate strategies using NDEA, Super-Efficiency Network Data Envelopment Analysis (SENDEA), Uncertain Network Data Envelopment Analysis (UNDEA), and Uncertain Super-Efficiency Network DEA (UnSENDEA) models, then aggregate results with the help of the super-efficiency CCR model to address both risky and non-risky scenarios (Figs. 2–7).

3. Problem statement

In this study, we aim to assess the effectiveness of blockchain

deployment strategies in the O&G industry to identify the most efficient approach. To accomplish this, we define evaluation criteria and establish relationships among them using NDEA models. Additionally, the evaluation models are enhanced to account for real-world conditions, including uncertainty and discrimination power.

3.1. Strategies and evaluation criteria

Following this methodology, our first step will be to identify strategies. Application novelty and the level of coordination are critical factors in defining blockchain deployment strategies. These strategies are summarized in Table 2 [36]. In the subsequent step, the strategy evaluation criteria, categorized under key factors encompassing supply chain, organization, technology, drivers, and pressures, are systematically compiled based on Munim et al. [36], as illustrated in Table 2.

3.2. Optimization models

The following sub-sections outline the architectural design and mathematical basis of our proposed framework for evaluating blockchain deployment strategies in the O&G sector. First, we present the NDEA approach that goes beyond traditional "black-box" models by incorporating the industry's two-stage structure of barriers, pressures, and drivers. Next, we develop Formulas (1)-(9) and show how the UnSENDEA model can be simplified into alternative versions (such as NDEA and SENDEA) for comparative evaluation. Lastly, a risk-sensitive aggregation method is offered based on super-efficiency CCR models to integrate findings across all models, providing practical insights for supply chain decision-makers in both high-risk and low-risk environments.

3.2.1. Architecture of the developed models

DEA serves as an efficient data analysis tool to discern best practices [40] and has been employed frequently in the energy industry [41]. In DEA, performance metrics are categorized as inputs and outputs. Multiple Decision-Making Units (DMUs) consume inputs to generate outputs. DEA models rely on optimization programming, constituting a fusion of data-oriented and optimization models. The whole-unit, or black-box, performance measurement is relatively straightforward, considering only the inputs and outputs of the DMU [42,43]. This simplicity enables a systematic expression of the model. In contrast, network system performance measurement is challenging due to various structures in the network production system. One such structure is the two-stage tandem system, where external inputs are used in the first stage to produce intermediate products, and subsequently utilized in the second stage to generate final outputs. It is crucial to note that the black-box concept may not offer an accurate representation and comprehensive overview of the actual existing system due to its disregard for existing connections. Here, we constructed network models to assess blockchain deployment strategies, considering supply chain, technological, and organizational barriers. These barriers, along with pressures from the market, customers, and partners, act as drivers for promoting the O&G industry.

Barriers represent the inputs of our evaluation system, while d serve drivers as its outputs. Both of these elements connect the system to the external environment, which is inherently full of uncertainty. The motivators and barriers, positioned as the intermediate layer of the evaluation system, emerge as a result of the interactions between inputs and outputs. An organization utilizing this evaluation framework is better equipped to recognize and assess the pressures arising from both ends (inputs and outputs) through this intermediate layer, rather than from the inputs and outputs alone. This perspective is illustrated in Fig. 2.

Fig. 2 guides the development of a network model where barriers lead to external pressures at the first level. Subsequently, these barriers and pressures contribute to the emergence of drivers in the second stage of the O&G industry. Using a two-stage-based system [43], we establish an NDEA model. This model, in contrast to typical network models, incorporates uncertainty and enhances discrimination power in evaluation. To address uncertainty, we introduce random changes in inputs and outputs based on Wen [44], considering variances in the data. Following Zhu [40], removing the observed DMU from the reference set is crucial for increasing discrimination power in our model to assess blockchain deployment strategies.

3.2.2. Mathematical formulations

The modeling framework introduced through Formulas (1) to (9) provides a comprehensive approach to evaluating blockchain deployment strategies by integrating efficiency measurement with network structure, super-efficiency principles, and uncertainty handling. The UnSENDEA model forms the core of this framework, represented by Formulas (1) to (5), which extends traditional DEA by incorporating a multi-stage transformation process where inputs are first converted into intermediate products before being transformed into final outputs. This layered structure reflects the complexity of real-world decision-making environments, particularly in supply chain contexts where performance evaluation must account for internal processes. To further enhance decision support, Formulas (6) to (9) offer a complementary aggregation mechanism that consolidates the results from multiple evaluation models; i.e., NDEA, SENDEA, UNDEA, and UnSENDEA, into a unified efficiency score.

Building upon the aforementioned concepts, we introduce the UnSENDEA model in Formulas (1)-(5), with corresponding notations detailed in Table 3. This table lists the notations used in our article, categorized by indices, decision variables, and parameters. All blockchain application strategies and their evaluation criteria are grouped under indices. The values of each strategy in the evaluation criteria are classified as parameters, while the efficiency scores of the strategies fall under decision variables.

Objective function (1) assesses the efficiency of DMUs (strategies). Constraint (2) expresses the relationship between the weighted sum of other DMU's inputs and the observed DMU's inputs, considering uncertainty conditions. Constraint (3) pertains to intermediate products, while Constraint (4) represents the relationship between the weighted sum of other DMU's outputs and the observed DMU's outputs under

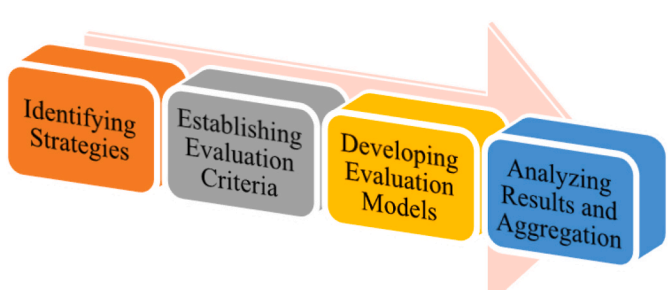

Fig. 1. Execution steps of the model.

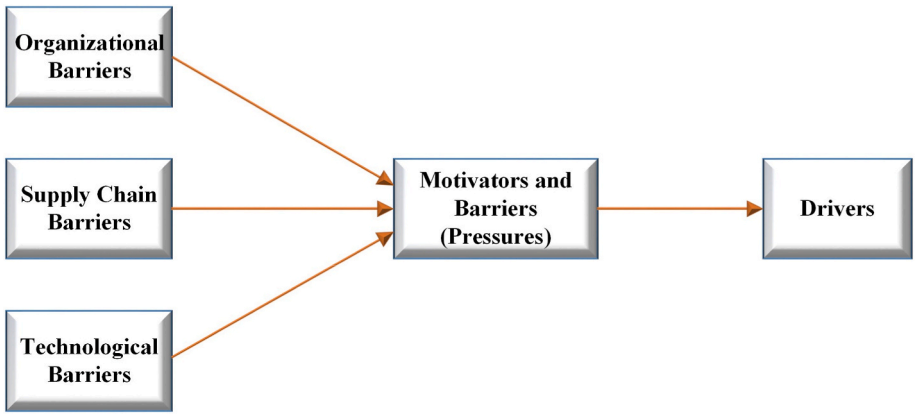

Fig. 2. Network structure evaluation in the O&G industry.

Table 3
Model notations.

Indices	Descriptions
$r \in R = \{1,\ldots,z\}$	DMUs
$i \in I = \{1,\ldots,m\}$	Inputs
$g \in G = \{1,\ldots,h\}$	Intermediate products
$j \in J = \{1,\ldots,s\}$	Outputs
$o \in R = \{1,\ldots,z\}$	DMUs under observation
Parameters	***Descriptions***
P_{ir}	Quantity of the ith input for the rth DMU
O_{jr}	Quantity of the jth output for the rth DMU
Z_{gr}	Quantity of the gth intermediate output for the rth DMU
ε	A minute positive value
σ_i, σ_j	Stochastic conditions based on the randomness of inputs and outputs according to the Standard Deviation (SD) of the data
$\varphi_{1-\alpha}^{-1}$	Fractile function where alpha indicates the chance of non-efficiency for each DMU
Decision Variables	***Descriptions***
θ	Efficiency level of each DMU
s_i^-, s_j^+	Slack and surplus variables
λ_r^1, λ_r^2	Peer weights for Stages 1 and 2

uncertainty. Constraint (5) defines the sign of the decision variables. Constraints (2)–(4) delineate distinct stages within the network evaluation process. Specifically, Constraint (2) establishes the link between external inputs and the internal workings of the evaluation system, representing the initial step (Step 1). Constraint (3) serves as the intermediary between Steps 1 and 2, while Constraint (4) signifies the connection between the internal processes of the evaluation system and the presentation of external outputs. In Constraint (5), the unit under observation is intentionally excluded from analysis, ensuring accurate super-efficiency calculations. Notably, Constraints (2) and (4) incorporate uncertainty terms, underscoring the model's consideration of uncertain conditions. Constraints (2) and (4), which both account for uncertainty conditions, apply such uncertainty across all dimensions of drivers and barriers (as stated in Table 2).

$$\text{minimize} \quad \theta - \varepsilon \left(\sum_i s_i^- + \sum_j s_j^+ \right) \tag{1}$$

$$\text{subject to} \quad \sum_{r \neq o} \lambda_r^1 P_{ir} + s_i^- = \theta \left(P_{io} + \sigma_i \varphi_{1-\alpha}^{-1} \right) \quad \forall i, \tag{2}$$

$$\sum_{r \neq o} \lambda_r^1 Z_{gr} \geq \sum_{r \neq o} \lambda_r^2 Z_{gr} \quad \forall g, \tag{3}$$

$$\sum_{r \neq o} \lambda_r^2 O_{jr} - s_j^+ = \left(O_{jo} + \sigma_j \varphi_{1-\alpha}^{-1} \right) \quad \forall j, \tag{4}$$

$$\lambda_r^1, \lambda_r^2, s_i^-, s_j^+ \geq 0, \theta \text{ free variable} \quad \forall i, j; r \neq o. \tag{5}$$

Further analyses of the presented model and its derivatives are possible. Removing uncertainty terms and the " $r \neq o$" term transforms the model into an NDEA model. Introducing the condition " $r \neq o$" to NDEA results in the SENDEA[1] model. The inclusion of uncertainty[2] terms in the NDEA model leads to the UNDEA model. Each of these models may yield distinct efficiency scores. After evaluating the four strategies outlined in Table 2 across the 15 criteria listed in Table 2 using NDEA, SENDEA, UNDEA, and UnSENDEA models, it is imperative to consolidate the assessment outcomes. We introduce both risky and non-risky scenarios for integration. Employing the super-efficiency CCR model (i.e., Formulas (6)-(9)), aligned with the results obtained from the mentioned models, integrates the evaluation outcomes. The DMUs are the blockchain deployment strategies, with inputs set uniformly in the non-risk case (SECCR-1) for all decision units. In the risky case (SECCR-2), inputs are equivalent to the SD of the strategy evaluation results across all evaluation models. A higher SD of efficiency scores indicates a riskier strategy selection, as there is no consensus on its efficiency value. Hence, Formulas (6)-(9) encapsulate the model's aggregation of evaluation results in both risky and non-risky scenarios.

$$\text{minimize} \quad \theta - \varepsilon \left(\sum_i s_i^- + \sum_j s_j^+ \right) \tag{6}$$

$$\text{subject to} \quad \sum_{r \neq o} \lambda_r P_{ir} + s_i^- = \theta P_{io} \quad \forall i, \tag{7}$$

$$\sum_{r \neq o} \lambda_r O_{jr} - s_j^+ = O_{jo} \quad \forall j, \tag{8}$$

$$\lambda_r, s_i^-, s_j^+ \geq 0, \theta \text{ free variable} \quad \forall i, j; r \neq o. \tag{9}$$

In essence, four data-driven optimal evaluation models have been devised via Formulas (1)-(5), utilizing super-efficiency, network

[1] **Super-Efficiency in DEA:** Super-efficiency refers to an enhanced DEA approach where the observed DMU is excluded from the reference set ($r \neq o$ condition) to enable efficiency scores above 1.0. This modification, as implemented in our SENDEA and UnSENDEA models, increases discriminatory power.

[2] **Stochastic Conditions in DEA:** Stochastic conditions in our network DEA framework incorporate data variability through

- Random adjustments to inputs/outputs based on their SDs,
- Fractile functions controlling the chance of non-efficiency.

structure, and uncertainty considerations. The network structure, depicted in Fig. 2, provides the foundation for these models. Subsequently, Formulas (6)-(9) facilitate the aggregation of evaluation outcomes derived from the aforementioned models, accommodating both risky and non-risky scenarios. This ensures supply chain managers encounter cohesive results, mitigating the challenge of disparate evaluation outcomes. Constraints (2)–(5) and Constraints (7)–(9) are all linear. All the models presented are based on linear programming, which ensures the ability to report globally optimal solutions. It is important to note that the constraints stem from DEA, which governs the relationships between inputs, outputs, and quasi-inputs/outputs.

3.3. Justifications and concepts behind our approach to problem-solving

This study offers a comprehensive approach to assessing the effectiveness of blockchain deployment strategies in the O&G industry. It integrates various methodologies and models to address the complexities inherent in assessing such strategies, particularly in the context of real-world conditions characterized by uncertainty and diverse performance metrics. One of the primary contributions of this work lies in the development of a novel decision-aid model, the UnSENDEA model. This model not only considers the efficiency of DMUs; i.e., blockchain deployment strategies in this case, but also incorporates uncertainty and enhances discrimination power in the evaluation process. By introducing random changes in observed inputs and outputs to address uncertainty and by removing the observed DMU from the reference set to enhance discrimination power, the UnSENDEA model provides a more nuanced and realistic assessment of strategy efficiency.

The integration of network structures into the evaluation process is another notable aspect of this study. Fig. 2 illustrates how barriers, external pressures, and drivers interact within the O&G industry, forming a two-stage network model that informs the evaluation framework. This approach recognizes the interconnectedness of factors influencing strategy effectiveness and provides a structured method for incorporating these dynamics into the evaluation process. Furthermore, the study outlines different variations of the evaluation model, including the NDEA, SENDEA, and UNDEA models, each offering distinct insights into strategy efficiency. By evaluating blockchain deployment strategies across these models and consolidating the outcomes using the super-efficiency CCR model, the study provides supply chain managers with a comprehensive understanding of strategy performance under both risky and non-risky scenarios.

The novel decision-aid model proposed in this study addresses a critical need in production research by offering a structured and network-based framework for designing and managing production systems and logistics. By employing advanced methodologies such as the UnSENDEA model, the research provides an operational tool for decision-makers to appraise the effectiveness of blockchain deployment strategies. This model not only considers traditional efficiency metrics but also incorporates uncertainty and network structures, ensuring a more comprehensive assessment of strategy performance. Through the systematic integration of evaluation criteria spanning supply chain, organizational, technological, and market-driven factors, the decision-aid model enables stakeholders to make informed decisions regarding strategy selection and resource allocation, ultimately enhancing the efficiency and resilience of production systems in the O&G industry. Moreover, the universality of the decision-aid model makes it accessible to a wide audience in production research.

By delineating the evaluation process through the offered methodology and illustrative framework, such as Fig. 2 depicting the network structure of the O&G industry, the model facilitates understanding and application across various organizational contexts. For instance, in this study, the O&G industry is examined because it is indeed considered a production system. It encompasses stages such as the extraction, refinement, and distribution of petroleum and natural gas resources, involving complex processes and logistical operations similar to those found in manufacturing and production sectors. Additionally, the flexibility of the model, demonstrated through its ability to accommodate both risky and non-risky scenarios, ensures its relevance and applicability in dynamic production environments. By integrating various models, considering uncertainty, and accounting for network structures, the study provides valuable insights that can inform decision-making and enhance the management of production systems and logistics in the industry. As a result, the decision-aid model presented in this study not only contributes to advancing knowledge in production research but also offers practical utility for industry practitioners seeking to optimize their operations and adapt to evolving market conditions.

It is noteworthy that our article is compared to Munim et al. [36] on blockchain strategies in the O&G supply chain, which stands out for its innovations. This research delves into evaluation through optimization and efficiency, providing decision-makers and managers with more precise insights for informed decision-making. Furthermore, we incorporate stochastic and risky conditions in decision-making, enhancing the accuracy of real-world decision-making processes. As a result, the developed approach incorporates linear programming, data optimization, and the application of stochastic concepts in optimization models. We assume that introducing random variations in inputs and outputs effectively captures the uncertainty inherent in real-world conditions. Additionally, the study presumes that the different evaluation models (NDEA, SENDEA, UNDEA, and UnSENDEA) each provide unique insights into strategy efficiency and are well-suited to capture various aspects of performance. This layered modeling approach, as outlined in Table 4, ensures a comprehensive and realistic assessment, enabling managers to choose strategies based not only on theoretical efficiency but also on real-world conditions.

Fig. 3 displays the proposed methodological framework.

4. Results

4.1. Case study

In this section, first, the data used to conduct the study is described. Next, the findings and analyses are presented to demonstrate the applicability and validity of the proposed methodology. Data[3] were sourced from Munim et al. [36] and obtained through a web survey distributed across diverse departments in four key Norwegian O&G companies: Aker BP, Equinor, BW Offshore, GE O&G Offshore, Schlumberger Limited, and DNV. The data are displayed in Appendix A.

Table 4
Model comparison for blockchain strategies.

NDEA	SENDEA
• Captures the interconnected stages of an evaluation process • Focuses on network structures (e.g., interactions between barriers, drivers, and external pressures).	• Enhances discriminatory power by allowing performance evaluation of efficient units beyond the frontier. • Removes the observed DMU from the reference set to avoid self-comparison bias.
UNDEA	**UnSENDEA**
• Introduces uncertainty modeling by incorporating uncertain changes in inputs and outputs. • Introduces stochastic elements (random variations in inputs/outputs) to simulate real-world uncertainty.	• Combines all features: Network structures, super-efficiency, and uncertainty. • Combines the strengths of UNDEA and SENDEA and handles uncertainty while also offering strong discrimination between strategies. • Provides the most realistic and precise evaluation.

[3] Tables 1–6 in Munim et al. [36] display the data.

Figure 3 displays the proposed methodological framework.

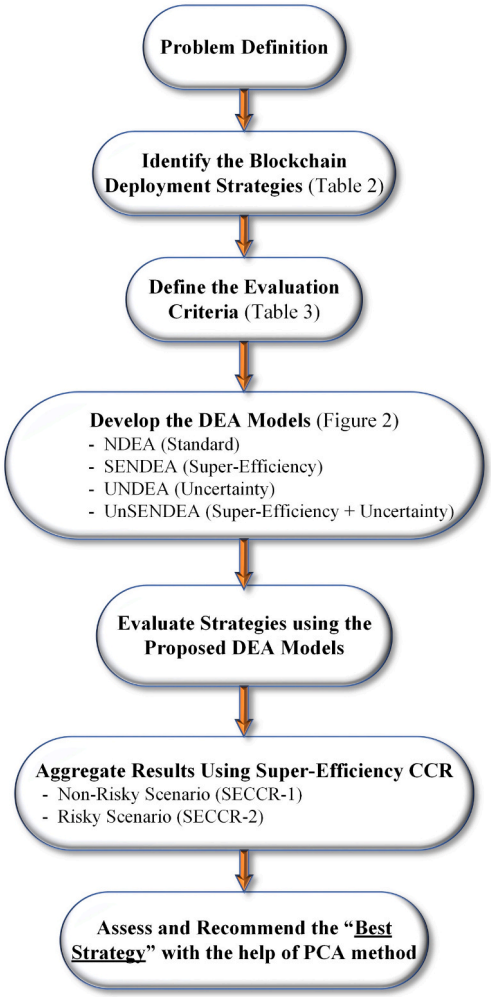

Fig. 3. Methodological framework.

The survey, available on Nettskjema. no (upon request), was disseminated through LinkedIn and email from April to October 2020. The sample comprises 10 respondents from Aker BP (four), Equinor (two), BW Offshore (two), and DNV (two). Of these, 2 possess Ph.D. degrees, 5 hold master's degrees, and 3 have Bachelor's degrees. Eight respondents work in private companies, while two are employed in public organizations. Further details can be explored in the first table presented by Munim et al. [36]. The prioritization of alternatives at the aggregate level, based on respondent opinions, is outlined in the last presented table via the same reference [36]. Utilizing this dataset, we assume σ_i and σ_j as random parameters within the 0 to 0.2 range, with α set at 0.05. We implement our models using the dataset adapted from Munim et al. [36].

4.2. Evaluation outcomes

Due to the integration of data-driven models and optimization techniques, our models do not necessitate substantial computational resources. They are consistently solved using GAMS software in under 0.1 s. The strategy evaluation results are presented in Table 5. The average efficiency scores for the single-use and substitution strategies are the highest. There is minimal difference in the average efficiency scores between the substitution and localization strategies. Models founded on super-efficiency exhibit higher discriminating power, as seen in SENDEA compared to NDEA. This means that SENDEA is more effective at distinguishing between the relative efficiencies of different DMUs, allowing for a more detailed evaluation of performance. By emphasizing super-efficiency, SENDEA can identify units that exceed the efficiency frontier, offering valuable insights into best practices and exceptional performance that standard efficiency models might overlook. The improved discriminating ability of SENDEA can aid decision-makers in making more informed decisions regarding resource allocation, performance improvement strategies, and benchmarking against best practices.

Table 5
Blockchain deployment strategies evaluation.

Strategies	NDEA	SENDEA	UNDEA	UnSENDEA
Single-use	1.000	1.451	0.763	1.057
Localization	0.931	0.937	0.610	0.610
Substitution	0.804	0.949	0.634	0.712
Transformation	0.862	0.871	0.608	0.611

Models incorporating uncertainty in the scores report lower efficiency, exemplified by UNDEA compared to NDEA. These variations attributable to the models underscore the importance of a comprehensive strategy evaluation to provide decision-makers and supply managers with accurate and comprehensive performance assessments. Fig. 4 illustrates the SD of efficiency scores generated by the evaluation models. A higher SD signifies greater distinctiveness in results, reflecting differences in efficiency scores among evaluation strategies. As shown in Fig. 4, super-efficiency-based methods, particularly SENDEA, exhibit the highest discriminatory power.

4.3. Results analysis

This sub-section interprets the evaluation outcomes from Section 4.2, focusing on rank consistency and uncertainty impacts.

4.3.1. Rank stability and strategic discrimination

In Fig. 5, the normalized rank distances across various evaluation methods are depicted. Notably, the SENDEA model exhibits the most substantial distance between Ranks 1 and 2, underscoring the robust placement of the single-use strategy in Rank 1. Conversely, the UnSENDEA model demonstrates the most significant difference between Ranks 2 and 3, further contributing to the diversity and robustness of the rankings. Additionally, SENDEA again displays the most noticeable gap between Ranks 3 and 4, emphasizing the distinctiveness of the evaluation outcomes. A higher difference signifies more robust rankings and the observed substantial gaps in SENDEA support the clear and secure position of the single-use strategy in Rank 1. It is noteworthy that Figs. 4 and 5 depict a comparison of the capabilities of various evaluation models. Larger gaps (like in SENDEA) indicate more stable rankings between top-performing and lower-performing strategies.

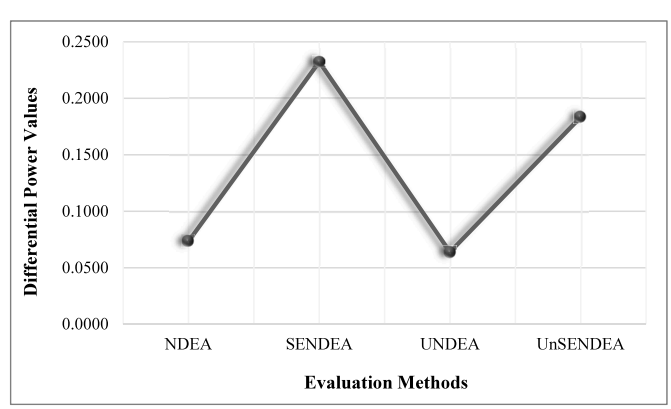

Fig. 4. SD of the efficiency scores for evaluation models.

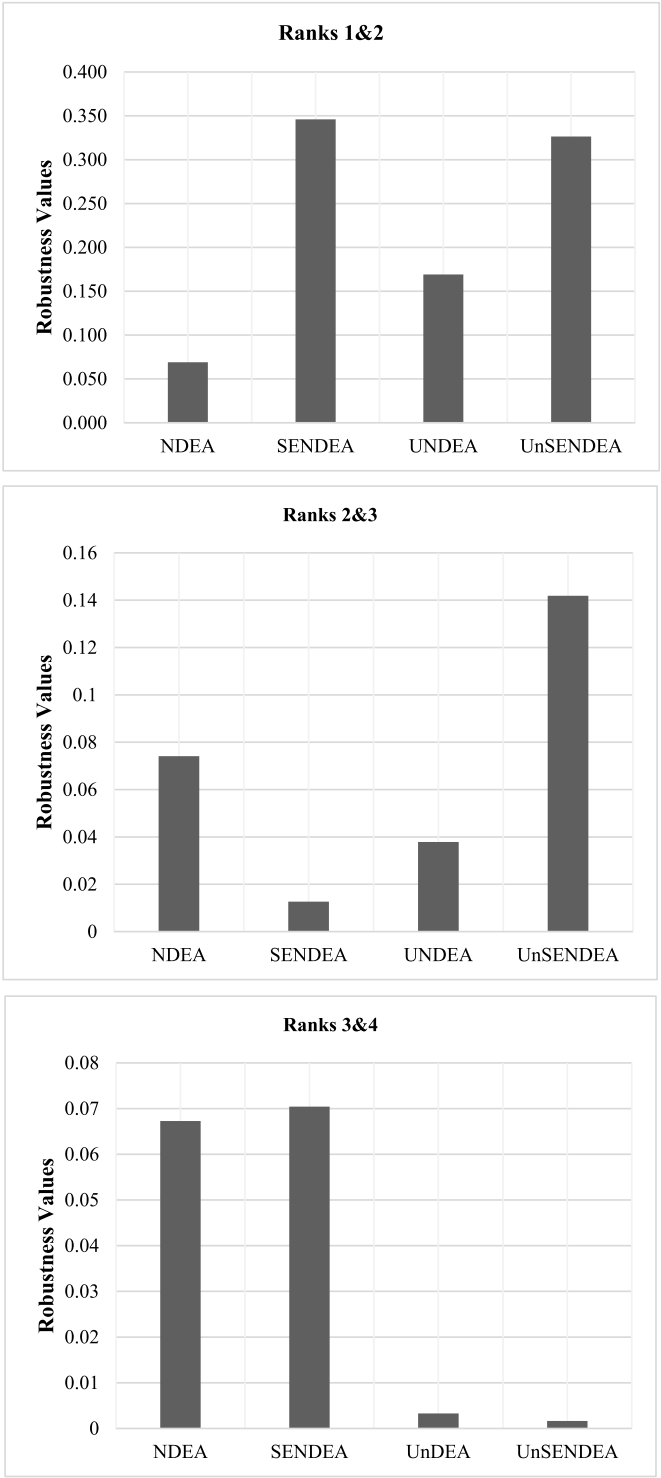

Fig. 5. Normalized rank distances.

4.3.2. *Risk-aware performance evaluation*

The evaluation outcomes are succinctly depicted in Fig. 6. The graph lines in this figure show strategy rankings with and without risk scenarios. In the high-risk scenario, the single-use strategy demonstrates variability, while the substitution strategy claims the top rank. This suggests that the single-use strategy encounters diverse efficiency values under high-risk conditions, lacking a strong consensus on its effectiveness. Employing such analyses to consolidate performance expectations proves beneficial for supply chain managers, allowing them to assess the consequences of choosing a strong strategy through a risk-focused lens during decision-making.

Uncertainty is inherent in decision-making processes, often accompanied by errors in data collection. Acknowledging and addressing uncertainty is crucial for supply chain managers to make informed decisions. Fig. 7 illustrates the effectiveness of blockchain application strategies, represented by the geometric mean, under both deterministic and non-deterministic conditions. This figure displays average efficiency scores under both deterministic and uncertain scenarios, enabling comparative analysis of real-world conditions.

Non-deterministic conditions exhibit lower efficiency compared to deterministic ones. This outcome aligns with the observation that uncertainty diminishes efficiency scores. Furthermore, while the single-use strategy remains relatively insensitive to uncertainty, the localization and substitution strategies demonstrate sensitivity to uncertain conditions.

4.3.3. *Confidence level trade-offs*

Fig. 8 depicts the evaluation outcomes considering network conditions, uncertainty, and super-efficiency at confidence levels of 99% and 95%. In other words, the bars in this figure illustrate how efficiency changes as the confidence level increases from 95% to 99%. As the confidence level increases, there is a corresponding reduction in the quality of results, as intended. Hence, elevating the confidence level while mitigating uncertainty leads to a decrease in the quality of efficiency values.

4.4. *Comparative analysis*

Two critical aspects of decision-making are the ability to differentiate between alternatives and the confidence in the chosen option. A decision-making method exhibits higher discriminatory power when it effectively distinguishes between alternatives. Conversely, a decision maker can confidently select an option when there is a significant difference in the score intervals of the top-ranked choices, particularly between Ranks 1 and 2. In light of this, we present our developed models alongside the latest work by Munim et al. [36] in Fig. 9. The bars in this figure represent concepts related to the SD and the distance between ranks. Our proposed models outperform existing methods in both discriminatory power and confidence in decision-making, as demonstrated by clearer discrimination between alternatives and greater score intervals between top-ranked options.

4.5. *ML application*

PCA is one of the most widely used multivariate statistical techniques across scientific disciplines [45]. The main objective of PCA is to extract the most important patterns of variation in the data and represent them using a smaller set of new, uncorrelated variables called *principal*

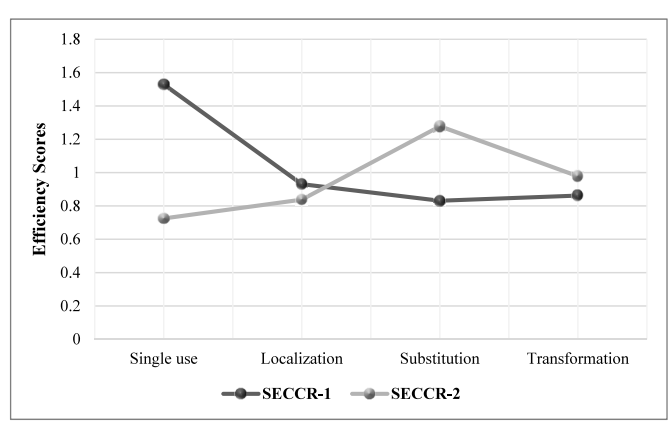

Fig. 6. Performance ranking and risk analysis.

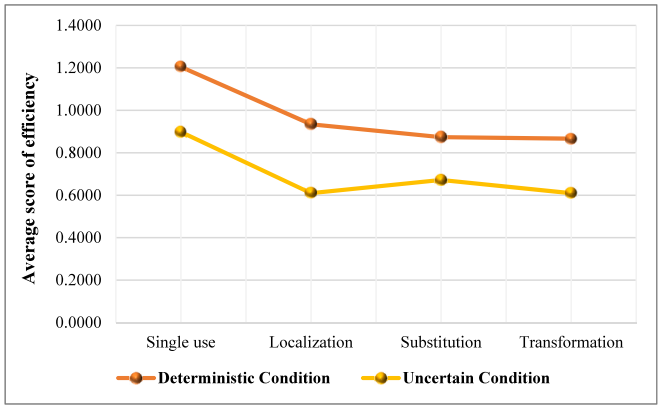

Fig. 7. Comparison of blockchain application strategies in both uncertain and deterministic scenarios.

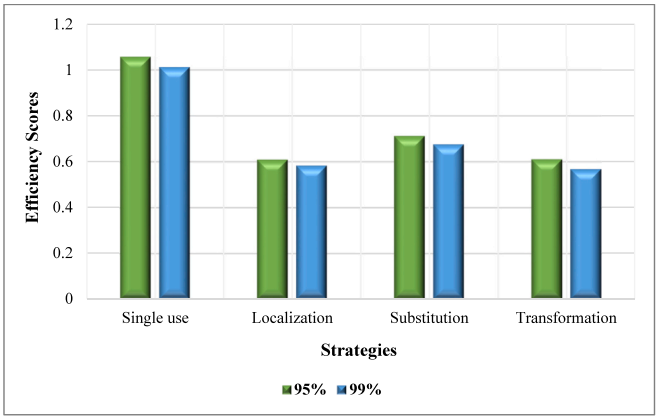

Fig. 8. Evaluation results at varying confidence levels.

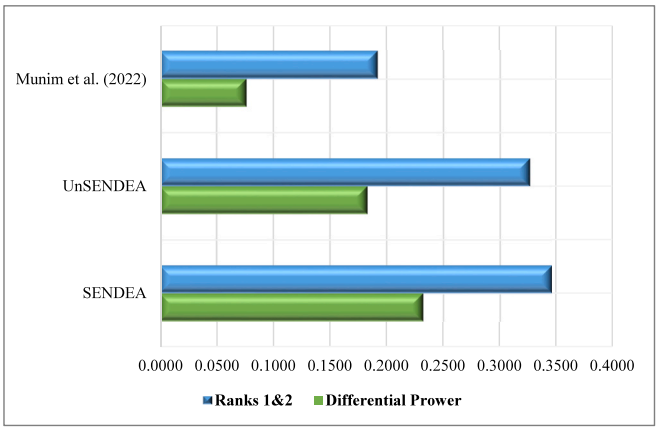

Fig. 9. Comparison of the decision-making models.

components. Given that the models in our study were all designed to evaluate strategies optimally, we employ PCA to explore the evaluation results in a data-driven way. In contrast to conventional datasets derived from observational or experimental data, our research conducts the analysis on optimized datasets generated through optimization models. This method provides decision-makers with insights that are not only data-driven but also based on optimal performance. PCA is applied to evaluate and visualize the relationships among the four investigated strategies in this research across the four evaluated methods in Table 5. Accordingly, PCA transforms the original data into two new axes called Principal Component 1 (PC1) and Principal Component 2 (PC2), which are derived from the patterns of variation across all evaluation methods. These components are linear combinations of the original features and are chosen such that PC1 captures the highest variance in the dataset, while PC2 captures the second-highest variance and is orthogonal to PC1. According to Fig. 10, the analysis reveals that PC1 explains approximately 89% of the total variance, indicating it accounts for the most significant underlying pattern in the data, whereas PC2 contributes an additional 11%, capturing secondary differences among the strategies.

According to Fig. 11, the PCA-transformed scores show that single-use strategy has the highest score on PC1 (3.255) and nearly neutral on PC2 (−0.042), suggesting that this strategy behaves distinctly different from the others along the primary axis of variation. In contrast, substitution strategy scores the lowest on PC1 (−1.054) and also exhibits the lowest value on PC2 (−0.947), placing it at the opposite end of the spectrum compared to the single-use strategy. Localization strategy scores moderately on PC1 (−0.816) but has the highest score on PC2 (0.892), highlighting its unique behavior along the secondary pattern. Transformation strategy represents the lowest score on PC1 (−1.385) and a near-zero score on PC2 (0.097), indicating it shares some similarities with substitution strategy but differs slightly in its secondary characteristics. These spatial relationships indicate that single-use strategy stands out as an outlier, while localization and transformation strategies show some degree of similarity. The explained variance ratio further supports the effectiveness of PCA in reducing dimensionality; together, PC1 and PC2 capture nearly 99.7% of the information present in the original four-dimensional dataset. Finally, single-use strategy is clearly different from the others which stands out as the most unique strategy. Substitution strategy behaves very differently from single-use strategy. Localization and transformation strategies are more similar to each other than to the other two strategies. These results are also consistent with Fig. 8, which displays that two strategies receive higher scores, while the other two represent similar performance.

In contrast to the previous analysis, now we perform PCA on the strategies. In other words, the PCA is applied to the dataset containing four features; i.e., single-use, localization, substitution, and transformation. The goal is to reduce dimensionality while preserving as much information as possible. The analysis reveals that two principal components; i.e., PC1 and PC2, are sufficient to capture nearly all of the variability in the original data. According to Fig. 12, PC1 alone accounts for 87.61% of the total variance, representing the main direction along which the strategies differ most significantly. In other words, this component captures the dominant underlying pattern in the data. When combined with PC2, which explains an additional 12.37% of the variance, the total amount of variation retained reaches 99.98%. This means that by reducing the data from four dimensions down to just two, we lose

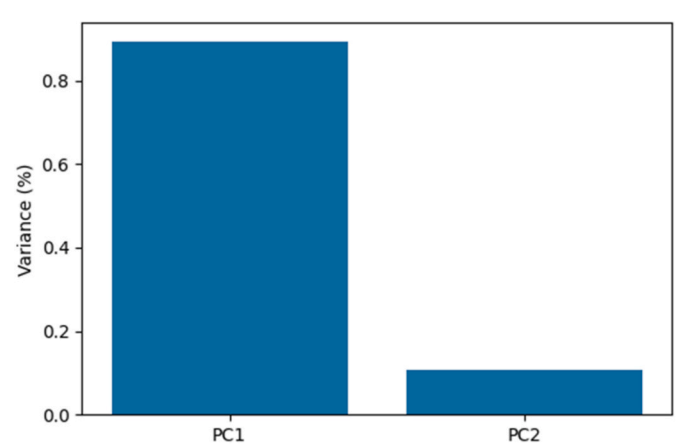

Fig. 10. Variance explained by the principal components.

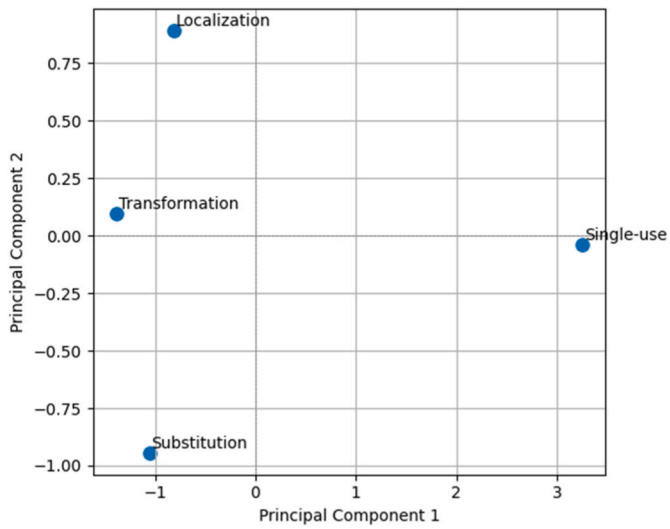

Fig. 11. PCA of strategies across the evaluation methods.

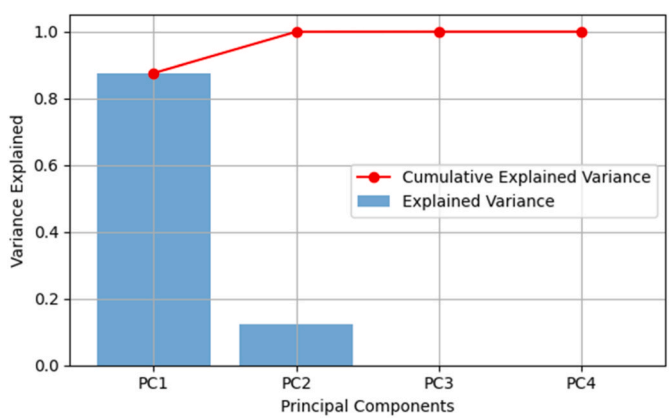

Fig. 12. PCA-explained variance ratio.

almost nothing in terms of the overall structure and relationships between the strategies.

Looking at Fig. 13, each evaluation method is projected onto this new two-dimensional space defined by PC1 and PC2. SENDEA has the highest score on PC1, making it the most distinct in terms of its feature values compared to the others. From a manager's standpoint, SENDEA is the most robust or widely applied method, possibly offering the broadest impact. UNDEA, on the other hand, has the lowest score on PC1, placing it far in the opposite direction. UNDEA lags behind in terms of overall usage or effectiveness across the board. NDEA and UnSENDEA occupy intermediate positions along PC1 but diverge more clearly along PC2, showing that they differ from one another in ways captured by this secondary axis. These results are consistent with the findings presented in Fig. 4.

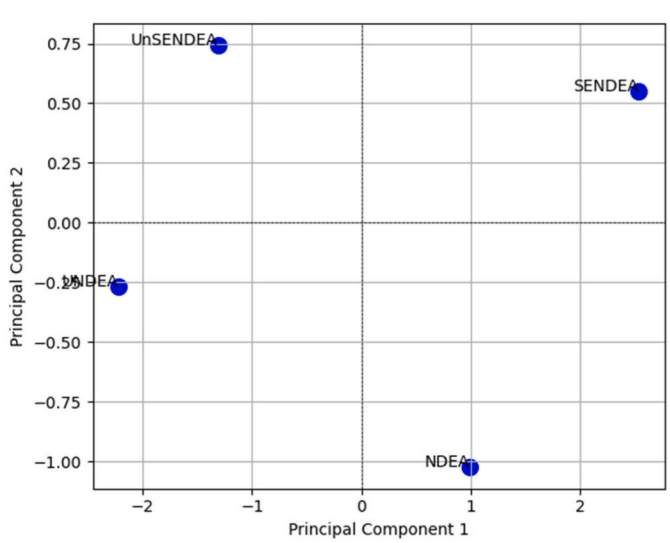

Fig. 13. PCA of the methods.

4.6. Discussion

This study draws attention to the key role that strategic decision-making plays in the successful integration of blockchain technology in the O&G sector. With the application of advanced analytical frameworks such as network DEA and a focus on uncertainty, supply chain managers can make strategic decisions in order to enhance operational efficiency and minimize risks. Specifically, our models are capable of evaluating the effectiveness of blockchain implementation strategies precisely using evaluation criteria and relevant data. These models are highly flexible and can be utilized across various sectors and problem types, independent of specific case contexts. Unlike typical data-based models whose outcomes largely depend on input data, our framework take advantage of mathematical optimization as its foundation. The purpose of the case study is to illustrate the models' performance numerically. The proposed models are built based on optimization logic and that is why they require significantly less data to perform data analysis.

Blockchain strategy evaluation within the O&G sector remains limited, with only three prominent studies to date; i.e., Munim et al. [36], Babaei et al. [39], and Babaei et al. [34]. Our research sets itself apart from these works in several significant ways.

➢ *Optimization Models Introduced*

Unlike Munim et al. [36], we incorporate linear programming-based optimization models in order to include a novel dimension to the evaluation.

➢ *Focus on Stochastic Conditions*

While Babaei et al. [39] addressed specific two-fold uncertainty conditions, our research emphasizes the analysis of stochastic environments.

➢ *Risk Calculation under Different Conditions*

By analyzing both stochastic and deterministic risk scenarios, and applying models with varying degrees of discrimination such as traditional and super-efficiency models, this work enhances understanding of blockchain strategy outcomes through a more detailed analysis. Unlike Babaei et al. [34], we incorporate network analysis based on real-world conditions, taking into account the interrelationships between entities such as criteria.

5. Managerial insights and practical implications

5.1. Blockchain integration

Across a range of sectors, including O&G and renewable energy, enterprises are increasingly utilizing blockchain technology to mitigate operational inefficiencies, enhance traceability, and address Environmental, Social, and Governance (ESG) imperatives, while simultaneously modernizing legacy infrastructure. Initially perceived as a niche innovation, blockchain has evolved into a transformative enabler within traditional industries, offering strategic benefits such as process

automation, heightened security, and considerable cost efficiencies [46]. In response to growing shareholder demands for cost efficiency and higher profits, the O&G sector is increasingly turning to blockchain to streamline its supply chains through enhanced transparency and automation.

Chevron Technology Ventures (CTV) was created in 1999 in order to incorporate external technologies aimed at enhancing Chevron's energy production and distribution, with a strategic emphasis on affordability, reliability, and environmental sustainability. Recently, Chevron partnered with Carbon Management Solutions (CMS) and Xpansiv on a blockchain pilot in Singapore in order to improve the traceability and auditability of lower-carbon fuel oil transactions via an asset ledger. CMS serves as a decarbonization platform within the commodities market, whereas Xpansiv offers the underlying infrastructure for the management and exchange of environmental commodities [47]. Similarly, Repsol [54], a leading Spanish energy company, collaborated with Finboot and its Technology Lab to employ blockchain technology for improving product certification. This partnership led to the creation of BlockLabs, a platform that digitizes sample approval processes and reduces inefficiencies like rework and communication delays (Repsol, Jan 29, 2024).

Beyond its origins in digital finance, blockchain demonstrates considerable potential across diverse domains, particularly in industrial SCM. Traceability plays an important role for companies seeking enhanced trust, the appreciation of sustainability initiatives, and effective communication with customers. Launched in 2018, The Energy Origin (TEO) employs blockchain technology, providing a blockchain-based solution to trace the origin of energy. The platform enables the creation of an auditable trail for certifying the origin and distinctive attributes of energy production [48]. In the emerging landscape of the grand transition to lower-carbon energy sources, there are broader implications. According to Forbes [49], it is anticipated that numerous industrial companies will adopt blockchain networks for the monitoring and tabulation of their carbon emissions. Pemex, a major global petroleum producer, has unveiled plans to integrate a blockchain technology platform for O&G SCM [50]. Joining the Petrobloq[4] Global Blockchain Industry Consortium, operated by Petroteq Energy Inc., marks a strategic move for Pemex. As a pioneer in accepting cryptocurrency payments, Pemex aims to optimize its extensive operations through blockchain adoption, enhancing efficiency from production to sales.

5.2. Model development

Considering blockchain's growing significance in the O&G supply chain, there is a notable gap in research on evaluating its application strategies within this industry. This study utilizes data-driven and optimization-based models for blockchain deployment strategies, considering innovativeness and complexity. Criteria encompass supply chain, organization, technology, and pressures. The developed evaluation models incorporate factors of uncertainty and discrimination that are crucial for supply chain decision-making. Our exploration of blockchain strategies in the Norwegian O&G sector, as depicted in Table 5, exhibits diverse outcomes based on super-efficiency and uncertainty conditions. Fig. 4 shows the superior discrimination power of super-efficient models in evaluating blockchain strategies. Fig. 5 reveals different robust capacities among the models, with super-efficient models demonstrating satisfactory robustness in ranking blockchain deployment strategies. In Fig. 6, despite the single-use strategy securing high scores across evaluation methods, it represents significant score dispersion. In summary, our study is delineated into two phases as illustrated in Fig. 14.

As can be seen in Fig. 14, the first phase entails evaluating approaches through various data-driven models. Subsequently, in the second phase, the evaluation findings are synthesized under risky and non-risky conditions, empowering supply chain managers to make informed and comprehensive decisions.

5.3. Insights and implications

Our approach, findings, and discussions provide critical insights into practical applications in production systems and logistics, with a focus on the O&G sector. The research integrates data-driven models and optimization strategies into a framework that is computationally efficient, making it well-suited for practical use in production and logistics environments. The quick solution times of less than 0.1 s achieved with GAMS software emphasize the practicality of applying these models in real-time operations. The evaluation results presented in Table 5 demonstrate the effectiveness of various strategies, including single-use, substitution, and localization, for optimizing production and logistics operations. These findings help decision-makers improve resource allocation and operational efficiency. On the other hand, this work tackles the challenge of uncertainty in decision-making, which is indisputable for dynamic production and logistics settings. By incorporating uncertainty factors and utilizing models like UnSENDEA, decision-makers can more accurately assess and manage risks, thus enhancing resilience and adaptability in their supply chains.

5.3.1. General managerial insights

The use of blockchain technology in the O&G supply chain, exemplified by Chevron's pilot initiative in Singapore and Repsol's partnership with Finboot, demonstrates the real-world applicability of our approach. Blockchain technology provides significant benefits in terms of improved transparency, traceability and operational efficiency, key components of contemporary production systems, and logistics management. The most important general managerial insights are given as follows.

i. **Strategy Effectiveness**: It is necessary to assess the performance of various approaches, including single-use, substitution, and localization, and identify the most effective methods for enhancing production and logistics operations,
ii. **Uncertainty**: Managers must acknowledge how uncertainty affects efficiency scores and focus on strategies that provide resilience and stability in unpredictable conditions,
iii. **Blockchain Solutions**: It is helpful to investigate how blockchain technology can be integrated into the supply chain to enhance transparency, traceability, and operational efficiency,
iv. **Resource Allocation**: Policymakers must implement evaluation models to strategically allocate resources, maximizing production and logistics effectiveness,
v. **Robust Evaluation Methods**: It is necessary to employ evaluation techniques to guarantee reliable and accurate decision-making.

5.3.2. General perspectives

On the other hand, our framework offers numerous valuable perspectives, as outlined below.

- *Integration of Emerging Technologies:* Our framework presents an opportunity to explore the integration of emerging technologies, such as advanced Artificial Intelligence (AI) and other emerging technologies, to enhance real-time efficiency measurement in production systems and logistics within the O&G industry,
- *Dynamic Decision Support Systems:* There is scope for creating decision support systems that dynamically respond to temporal variations in environmental conditions and market dynamics,

[4] Petrobloq stands as the pioneering blockchain platform exclusively crafted to address the intricate supply chain requirements of the Oil & Gas sector.

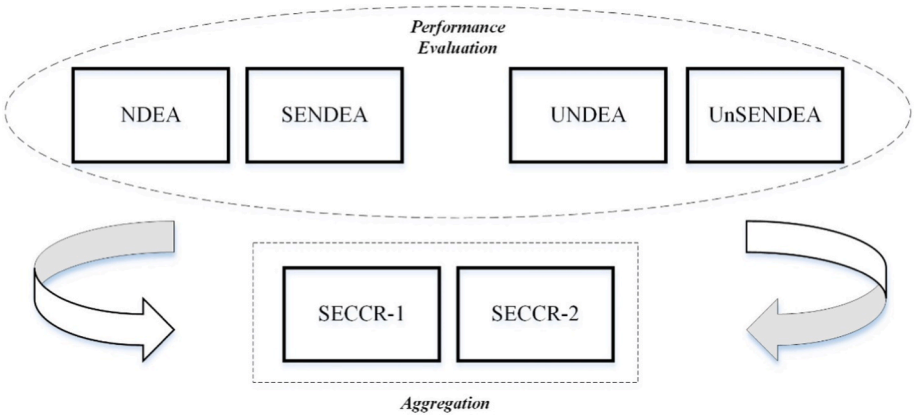

Fig. 14. Two-phase evaluation framework for the supply chain strategies.

- *Cross-Industry Application:* Although our research primarily addresses the O&G sector, the models presented could be extended to industries encountering similar supply chain complexities.

This work has the potential to extend the scope of evaluation for blockchain adoption in other contexts, such as renewable energy and smart grids. This extension is particularly relevant given the growing emphasis on sustainability and the global shift toward technologies that prioritize traceability, accountability, and transparency. In renewable energy systems and smart grids, similar data-driven models, including DEA and its network variants, can be applied to assess performance, optimize operations, and manage uncertainties within the framework of blockchain adoption. It is important to note that our models are general and can be adapted to these contexts by adjusting the criteria and alternatives to suit specific requirements. Utilizing our models in sustainability-related industries enable decision-makers to develop more effective planning strategies, handle risks, and adapt to market changes, contributing to the development of sustainable energy frameworks.

5.3.3. *Practical application of results in the Norwegian O&G industry*

The findings of this study offer several practical implications for decision-makers, supply chain managers, and policymakers in Norway's O&G sector.

- **Informed Strategy Selection:** The study shows that single-use performs best under various evaluation models, especially in low-risk environments. Norwegian O&G firms can prioritize these strategies during the early phases of blockchain adoption, where return on investment and clarity of outcomes are critical.
- **Risk-Aware Decision-Making:** In high-risk scenarios, substitution outperforms single-use, suggesting its resilience for projects in the risk-intensive environments.
- **Model-Based Decision Support:** The proposed two-phase framework (Fig. 14) enables companies to:
  - o Assess strategies using data-driven models (e.g., SENDEA, UNDEA),
  - o Utilize PCA method to visualize how different strategies and evaluation methods relate to each other,
  - o Synthesize results under risky vs. non-risky conditions.
- **Policy and Collaboration:** Norway's government/industry bodies (e.g., the Norwegian petroleum directorate) should integrate the study's models into policy tools to standardize evaluations.
- **Operational Adjustments**: **Confidence-level trade-offs** include:
  - o Using 95% confidence for routine decisions,
  - o Applying 99% confidence for high-impact investments.

6. Conclusion and outlook

Adopting blockchain technology can greatly enhance the O&G SCM by improving transparency, security, transaction speed, and tracking accuracy. As a result, industry managers must assess blockchain deployment strategies to identify the most effective ones. In the following, major findings, policy suggestions and outlook for future studies are discussed.

6.1. *Major findings*

This research offered data-driven optimization models for evaluating these strategies. The assessment included single-use, localization, substitution, and transformation strategies based on criteria related to drivers, supply chain, organization, and technology. Viewing the criteria as a network revealed the interconnectedness, where overcoming obstacles incentivizes blockchain use in the O&G industry. Employing four NDEA models and considering non-determinism and super-efficiency conditions, we evaluated the Norwegian O&G industry. PCA technique was then applied to convert optimal data into a more interpretable form, preserving the most significant variation patterns. The findings highlighted the significant discrimination power of the SENDEA method. Notably, single-use and substitution strategies exhibited favourable efficiency for deployment in the industry.

Eventually, the following key points summarize the main findings.

I. **Efficiency of Models:** All models were efficiently solved using GAMS software in under 0.1 s, requiring minimal computational resources (Table 5).
II. **Top Strategies:** The highest average efficiency scores were observed for single-use and substitution strategies, with minimal differences between substitution and localization (Table 5).
III. **Super-Efficiency Models:** Super-efficiency models like SENDEA exhibited greater discriminating power compared to NDEA (Figs. 4, 5 and 13).
IV. **Impact of Uncertainty:** Models incorporating uncertainty, such as UNDEA, obtained lower efficiency scores than those without uncertainty (NDEA) (Fig. 7).
V. **Discriminatory Power:** SENDEA revealed the largest gaps between rankings, particularly between Ranks 1 and 2, demonstrating strong ranking robustness for the single-use strategy (Fig. 5).
VI. **High-Risk Scenario:** Under high-risk conditions, the substitution strategy ranked highest, while the single-use strategy displayed more variability in efficiency (Fig. 6).
VII. **Effect of Confidence Levels:** Increasing confidence levels (from 95% to 99%) reduced the quality of results due to the trade-off with uncertainty mitigation (Fig. 8).

6.2. Policy suggestions

This work provided a strong evidence supporting the strategic use of blockchain technology in the O&G sector, highlighting the importance of data-driven decisions and evaluation frameworks. First, prioritizing the creation of a comprehensive blockchain strategy at the national or industry level is essential. This strategy should align with specific requirements of the O&G sector, goals of SCM, technological integration, and organizational aspects. According to the findings, strategies like single-use and substitution achieve the highest operational efficiency, making them suitable for early-stage adoption. Risk management and decision-support systems can integrate advanced analytical models like the UnSENDEA framework proposed in the study. These models, which include super-efficiency and uncertainty considerations, provide a robust basis for evaluating blockchain strategies. The outcomes further encourage collaboration across industries and the integration of blockchain frameworks into renewable energy and smart grid infrastructures. This development mirrors global priorities in sustainability, alongside the intensified emphasis on tracking and responsibility. In other words, using the same assessment frameworks from the O&G industry in renewable energy supply chains helps ensure methodological consistency and encourages innovative solutions. On the other hand, the generalizability of our proposed models makes them adaptable to diverse contexts, which supports broader applications in the energy sector. In this regard, industry-level benchmarks for efficiency must be developed to enable meaningful comparisons of blockchain adoption among organizations. The findings on confidence levels (95% vs. 99%) emphasize the importance of flexible decision-making frameworks that effectively strike a balance between efficiency with uncertainty.

6.3. Future directions

Future research may focus on developing additional DEA models, applying our evaluation models in O&G companies, and adapting them for application in other industries. Doing so enables researchers to assess the models' real-world effectiveness and gain meaningful understanding of their practical use and impact. Given the relevance of blockchain technology across different sectors, future studies could explore the adaptation of the evaluation models for application in other industries facing similar challenges and opportunities. This analysis across industries could uncover the extent to which blockchain adoption strategies are transferable and the implications for sectors such as manufacturing, logistics, and healthcare. For example, the application of the developed UnSENDEA and SENDEA models can be taken into account across various geographical regions through collaboration with reputable O&G companies. This would allow researchers and practitioners to validate the effectiveness of these models under varying regulatory, environmental, and operational conditions. In particular, it would provide an opportunity to assess how well the single-use and substitution strategies, identified as top performers in this study, produce significant efficiency improvements when adopted widely. On the other hand, future research could benefit from applying fuzzy logic [51, 52] and robust optimization [53] to model uncertainties in data and address the inherent variability. This form of uncertainty allows for a more efficient treatment of qualitative data; e.g., expert evaluations or stakeholder opinions, thereby improving the models' adaptability.

CRediT authorship contribution statement

Ardavan Babaei: Writing – original draft, Conceptualization, Methodology, Data curation, Software. **Erfan Babaee Tirkolaee:** Writing – review & editing, Validation, Supervision, Investigation. **Shahryar Sorooshian:** Writing – review & editing, Formal analysis, Investigation. **Sadia Samar Ali:** Writing – review & editing, Formal analysis, Validation. **Gongming Wang:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Data

A1. Criteria and Alternatives

- Criteria include:
  1. LE (Lack of Expertise in Technology)
  2. LT (Lack of Tools for Blockchain Technology Implementation)
  3. LB (Lack of Benchmarking Data for Blockchain Technology Implementation)
  4. LCA (Lack of Customer Awareness about Blockchain Technology)
  5. LSCB (Lack of Supply Chain Partner Collaboration)
  6. LSCO (Lack of Supply Chain Partner Coordination)
  7. IT (Immaturity of the Technology)
  8. LI (Limited Information Technology Infrastructure) -
  9. SC (Security Concerns)
- Alternatives include:
  1. Single-use
  2. Localization
  3. Substitution
  4. Transformation

Table A1
Inputs.

P_{ir}	1	2	3	4	5	6	7	8	9
1	0.069	0.047	0.052	0.049	0.066	0.037	0.04	0.042	0.039
2	0.073	0.044	0.053	0.049	0.063	0.043	0.045	0.054	0.048
3	0.086	0.056	0.056	0.048	0.07	0.042	0.05	0.065	0.048
4	0.089	0.058	0.058	0.052	0.08	0.049	0.052	0.063	0.055

A2. *Criteria include:*

1. IIS (Increases in Information Security)
2. ROC (Reducing Operations Cost)
3. IIT (Increases in Information Traceability)

Table A2
Outputs.

O_{jr}	1	2	3
1	0.057	0.061	0.046
2	0.061	0.066	0.039
3	0.067	0.068	0.047
4	0.066	0.076	0.045

A3. *Criteria include:*

1. NCS (Need for Collaborating with Supply Chain Partners)
2. CP (Customer Pressures)
3. MP (Market Pressures)

Table A3
Intermediate products.

Z_{gr}	1	2	3
1	0.039	0.027	0.033
2	0.045	0.032	0.043
3	0.057	0.04	0.055
4	0.061	0.039	0.054

Data availability

Data will be made available on request.

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